MC 501 Data Wrangling in R rubric

50 points. Due Monday, November 16, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 50. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Length: one script of 50 to 150 lines, comments included, with each of the three reviewer-facing comments 40 to 100 words. Word counts cover your own writing, not the title, headings, tables, references, figure captions, code, or output, and writing more than the range never costs points. Scores on work that shows effort are adjusted with the course's curve: your final percentage is 25 percent plus 75 percent of your rubric percentage, so 50 of 100 on the rubric becomes 62.5. Work at two-thirds or more of the minimum length counts as showing effort. Attempts are unlimited, so you can revise an early submission right up to the deadline.

- Inferencing Data needs all four kinds of variable from this table: two categorical, one with three or more groups, and two continuous. Step 5 is where they are kept.
- What it feeds: Describing Data and Inferencing Data both read `twitch_analysis.RDS`. Your White Paper reuses the moves in step 1 (read a coding form in, make the codes factors, join by id) on your own data; the Twitch table itself stays here.
- Code the AURA Lab R Companion writes into your script counts as script code. Keep the comment line it writes above each block, and be ready to explain every line.
- The comments the R Companion writes describe what the code does and do not count toward the three reviewer-facing comments.
- Before you submit: restart R, run the script top to bottom, and see what breaks.

Data Wrangling in R, 50 points

Criterion	Exemplary	Proficient	Developing	Beginning
1. Hand-code, and read your codes back 4 points	**4 points.** One variable hand-coded for a 30-message sample drawn with `sample_messages()`, recorded in a coding-form spreadsheet, read back with `read_csv()` and `clean_names()`, made a labeled factor with `factor()`, and joined to the sample by message id.	**3 points.** Hand-coded and read back, but the codes are not made a factor, or not drawn with `sample_messages()` or not 30 messages.	**2 points.** Codes typed into the script instead of read from a coding form, or described but not saved.	**0 to 1 points.** No hand-coding.
2. Load 4 points	**4 points.** The chat and stream tables loaded inside the script.	**3 points.** Loaded from a file on your own computer, so the script fails anywhere else.	**2 points.** Only one table loaded.	**0 to 1 points.** Not loaded in the script.
3. Timestamps 4 points	**4 points.** Converted from epoch milliseconds to a date-time with `clean_dates()`, which divides by 1,000, or by hand, dividing by 1,000 first.	**3 points.** Converted, with a small error left, such as the time zone.	**2 points.** Converted without dividing by 1,000.	**0 to 1 points.** Not converted.
4. Message length 4 points	**4 points.** A new column holding each message's length.	**3 points.** Computed on the wrong field.	**2 points.** Computed but not saved as a column.	**0 to 1 points.** Missing.

Criterion	Exemplary	Proficient	Developing	Beginning
5. Join, and keep the analysis columns 4 points	**4 points.** Stream information joined on, with a comment naming the key, keeping the channel's main category (main_game), `is_gaming`, `viewer_band`, and one continuous audience measure (`median_viewers`, or each message's viewers from the snapshot nearest before it, which `join_chat_streams()` attaches).	**3 points.** Joined and the columns kept, but no comment names the key, or the continuous measure is missing.	**2 points.** The join matches nothing or multiplies the rows, unnoticed; or only `is_gaming` survives, so the table supports a t-test and nothing else.	**0 to 1 points.** No join.
6. Missing values 4 points	**4 points.** The count reported, and the decision stated.	**3 points.** Handled, but the count is not reported.	**2 points.** Dropped without saying how many.	**0 to 1 points.** Not handled.
7. Export 4 points	**4 points.** Saved as `twitch_analysis.RDS`.	**3 points.** Saved under another name.	**2 points.** Saved in a format that loses the column types.	**0 to 1 points.** Not exported.
8. Three reviewer-facing comments 15 points	**15 points.** Three comments, each naming a decision a reviewer could challenge, the rule you applied, and why.	**11 to 14 points.** Three comments, but one states the rule without the reason, or names a decision nobody would question.	**8 to 10 points.** One or two comments, or comments that describe what the code does instead of a decision you made.	**0 to 7 points.** None.
9. Clean-session run 7 points	**7 points.** Runs start to finish after restarting R.	**5 to 6 points.** Runs after one small fix, such as a missing `library()` line.	**4 points.** Stops partway in a fresh session.	**0 to 3 points.** Runs only with objects already in your environment, or not at all.